

UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

Phase 1: Brainstorming & Ideation — Idea Prioritization Matrix

Project Name: Cab Booking (`UCab`)
Project ID: `N/A (Solo Track Submission)`
Team Member / Solo Developer: Shaik Sumiya Zainab

1. Idea Brainstorming Session Summary

During the initial conceptualization of UCab, over 15 distinct features and architectural ideas were brainstormed to create a competitive full-stack cab booking platform. Since this project was engineered completely from scratch by a single developer, strict prioritization frameworks (Impact vs. Feasibility / MoSCoW Method) were applied to ensure timely, high-quality delivery without compromising on user delight or system reliability.

2. MoSCoW Prioritization Framework

Must Have (Core MVP & Evaluation Essentials)

- JWT Authentication & Role Gating: Secure token-based access separating standard riders from executive administrators (`protect` and `adminOnly` middleware).
- Multi-Category Fleet Discovery: Visual cab listing across 4 vehicle classes (Mini, Sedan, SUV, Luxury) with real-time sorting.
- Deterministic Fare Calculator: Automatic route distance (`km`) and price calculation (`Base Fare + Distance \* Price/KM`).
- Zero-Config Dual-Engine Storage: Automatic failover between Mongoose ORM and `data.json` atomic file storage to guarantee 100% demo success across all evaluation machines.

Should Have (User Delight & Differentiators)

- Interactive Live GPS Tracking Bar: Visual step-by-step progress tracker indicating driver assignment, cab arrival, transit, and ride completion.
- Promo Code Discount Engine: Live validation for codes `UCAB20` (20% discount) and `WELCOME10` (10% discount).
- In-Ride Refreshments Package (`+ 150`): Option to order chilled drinking water and snacks upon booking.
- Road Safety Charity Contribution (`+ 120`): Optional checkout donation to driver welfare funds.

Could Have (Added Polish & Documentation)

- Corporate Tax Invoice Modal (`ReceiptModal.jsx`): Printable thermal/corporate invoice with embedded QR code verification.
- 1-Click Demo Login Buttons: Instant demo account login for both riders and super admins to streamline evaluation testing.

Won't Have (Deferred for Phase 2 Scalability)

- Live Bi-Directional WebSocket Streaming: Replaced with simulated high-fidelity state tracking for Phase 1.
- Third-Party Payment Gateway Webhooks (Stripe/Razorpay): Replaced with automated saved payment simulation.

3. Impact vs. Feasibility Evaluation Matrix

Proposed Feature	User / Business Impact	Solo Dev Implementation Effort	Priority	Action
Tabbed JWT Authentication & Bcrypt Hashing	High	Medium (3 days)	**Executed (Core)**	
Zero-Config Mongoose + JSON Fallback Engine	High	High (2 days)	**Executed (Core)**	
Dynamic Fare & Distance Estimation Engine	High	Medium (2 days)	**Executed (Core)**	

****Multi-Class Cab Inventory (CRUD Admin)** High Medium (2 days) **Executed (Core)****
****Promo Code Validation & Perks Add-ons** Medium Low (1 day) **Executed (Differentiator)****
****Live Dispatch Status Bar (`UserHome.jsx`)** High Medium (2 days) **Executed (Core)****
****Printable PDF Corporate Invoice + QR Code** Medium Low (1 day) **Executed (Polish)****